

Chapter 3: Whole numbers

12 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-CAL-C-01 compares, orders and calculates with integers, and MA4-IND-C-01 (index notation, primes, factors and multiples).

CHAPTER

3

3.01 Rounding and estimation

3.02 Multiplication

3.03 Division

3.04 Divisibility tests

3.05 Long division

3.06 Roman numerals

3.07 Index notation and powers

3.08 Square roots and cube roots

3.09 Prime and composite numbers

3.10 Prime factors

3.11 Highest common factor

3.12 Lowest common multiple

How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes box**.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

DRILL **Skill drill pages.** After the Easy and the Medium examples there is a full page of short practice questions. Work down the page, write answers on the lines, then mark it and record your score and time.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

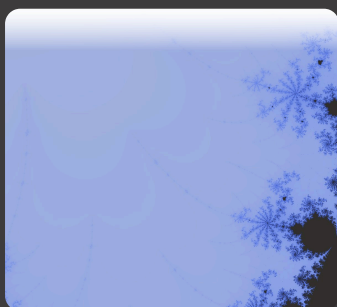
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
3.01	Rounding and estimation	3		☐
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3.12	Lowest common multiple	71		☐

In this chapter you will

- round and estimate
- multiply and divide whole numbers
- use divisibility tests
- use long division and Roman numerals
- use index notation, square roots and cube roots
- find prime numbers and prime factors
- find the HCF and LCM of two numbers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the tentacled edge of a small copy of the whole set, out on the antenna near -1.77 . Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
3.01

Rounding and estimation

Date _____

Learning intentions

We are learning to:

- round whole numbers
- estimate answers by rounding first

Success criteria

I can:

- ☐ find the **rounding digit** and look at the digit after it
- ☐ round to the nearest 10, 100, 1000 or 10 000
- ☐ estimate by rounding each number first

Key terms Write the definition as your teacher explains it.

Round

Estimate

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Round 87 531 to the place given.

a Round 87 531 to the nearest thousand.

b Round 87 531 to the nearest ten.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Round each number to the nearest ten.

12 QUESTIONS

REMEMBER Find the rounding digit.

1 47

2 152

3 3875

4 995

5 6004

6 87 531

Questions 7–12: round to the nearest hundred.

7 348

8 1250

9 7963

10 15 049

11 99 950

12 608

SKILL DRILL 1 Easy basics: practise until it is automatic

37 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 37

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–21: round to the place given.

1 6 358 (nearest 100)

2 2 694 (nearest 100)

3 383 (nearest 10)

4 4 546 (nearest 100)

5 627 (nearest 10)

6 62 (nearest 10)

7 98 (nearest 10)

8 14 (nearest 10)

9 4 902 (nearest 100)

10 4 655 (nearest 100)

11 71 539 (nearest 1 000)

12 4 816 (nearest 100)

13 221 (nearest 100)

14 259 (nearest 10)

15 43 791 (nearest 1 000)

16 971 (nearest 10)

17 563 (nearest 10)

18 8 838 (nearest 1 000)

19 482 (nearest 10)

20 5 935 (nearest 100)

21 12 691 (nearest 1 000)

ROUND 2 Questions 22–37: estimate: round each number to the nearest 10 first.

22 95 + 68

23 73 + 31

24 24 + 91

25 99 + 64

26 78 + 77

27 71 + 60

28 82 + 88

29 91 – 69

30 18 + 21

31 89 – 22

32 63 + 41

33 91 + 64

34 24 + 44

35 23 + 47

36 45 + 62

37 23 + 48

YOU DO

On your own: **Easy** Set B. Round to the place shown.

6 QUESTIONS

1 748 (hundred)

4 87 531 (ten thousand)

2 87 531 (hundred)

5 65 250 (thousand)

3 3499 (thousand)

6 149 999 (ten thousand)



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Estimate by rounding each number first.

a Estimate $425 + 81 + 940 + 22$.

Your teacher works this out. Copy every step.

b Estimate $864 + 250 - 417$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Estimate by rounding each number to its first digit.

6 QUESTIONS

REMEMBER Look at the digit to its right.

1 $612 + 285$

3 $38 + 71 + 405$

5 $7210 + 2890$

2 $4190 - 1820$

4 $783 - 219 + 96$

6 $5125 - 3980$

SKILL DRILL 2 Medium basics: keep practising

35 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 35

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: round to the first (leading) digit.

1 566

2 35 003

3 2 806

4 367

5 94 001

6 77

7 9 201

8 47 001

9 79 002

10 446

11 574

12 57

13 70

14 91 002

15 20

16 60

17 4 309

18 4 909

19 82 005

20 6 703

ROUND 2 Questions 21–35: estimate: round each number to its leading digit first.

21 84×194

22 48×617

23 29×202

24 53×963

25 40×148

26 88×245

27 21×145

28 37×649

29 56×758

30 34×295

31 62×610

32 66×560

33 97×494

34 73×48

35 56×604

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Estimate by rounding each number first.

a Estimate 93×19 .

Your teacher works this out. Copy every step.

b Estimate $2402 \div 32$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Estimate, then check with a calculator.

4 QUESTIONS

REMEMBER 5 or more: round up. 4 or less: stay.

1 48×21

3 $3950 \div 38$

2 612×9

4 A school has 28 classes of 31 students. Estimate the number of students.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Round 6382 to the nearest hundred.

Level 2

Estimate $512 + 289 - 97$.

Level 3

Estimate $4180 \div 21$.

 Extension · Stretch yourself Worked example

A number rounded to the nearest hundred is 4700. What are the smallest and largest whole numbers it could be?

1. Numbers from 4650 round up to 4700
2. 4750 rounds up to 4800, so stop at 4749

Answer: 4650 and 4749

YOU DO

On your own: read the worked example, then try these.

- E1** A number rounded to the nearest thousand is 12 000. Find the smallest and largest whole numbers it could be.

- E2** A crowd was “about 35 000” to the nearest thousand. What is the largest the crowd could have been?

Lesson summary

Missed the lesson? Start here.

Remember

1. Find the rounding digit.
2. Look at the digit to its right.
3. 5 or more: round up. 4 or less: stay.
4. Estimate: round first, then calculate.

 Level 1

Round 87 531 to the nearest thousand

1. Thousands digit 7, next digit 5
2. Round up

Answer: 88 000

 Level 2

Round 3499 to the nearest thousand

1. Next digit 4

Answer: 3000

 Level 3

Estimate 93×19

1. $\approx 90 \times 20$

Answer: 1800

 Extension

Nearest hundred is 4700

1. 4650 to 4749

Answer: 4650 and 4749

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LESSON
3.02

Multiplication

Date _____

Learning intentions

We are learning to:

- multiply whole numbers mentally and on paper

Success criteria

I can:

- ☐ use mental strategies such as doubling or splitting a number
- ☐ multiply by a one-digit number on paper
- ☐ multiply by a two-digit number on paper

Key terms Write the definition as your teacher explains it.

Product

Algorithm

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate mentally.

a 12×4

Your teacher works this out. Copy every step.

b 25×8 (use doubling)

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each product mentally.

14 QUESTIONS

REMEMBER Double and halve: $25 \times 8 = 50 \times 4$.

1 12×4

2 15×6

3 20×9

4 11×8

5 25×4

6 50×7

Questions 7–14: evaluate mentally.

7 7×8

8 9×6

9 12×5

10 40×3

11 11×12

12 25×6

13 6×70

14 8×125

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–28: know your times tables.

1 7×6

2 8×10

3 4×4

4 9×11

5 10×8

6 4×2

7 4×5

8 3×5

9 5×6

10 9×9

11 10×11

12 10×9

13 2×9

14 6×4

15 8×8

16 9×8

17 7×12

18 3×11

19 2×4

20 4×10

21 3×8

22 12×3

23 10×10

24 11×11

25 8×4

26 11×5

27 4×6

28 6×12

ROUND 2 Questions 29–44: multiply by 10, 100 or 1000: move the digits.

29 667×100

30 707×100

31 $234 \times 1\,000$

32 517×10

33 445×100

34 874×10

35 370×100

36 924×100

37 60×10

38 471×10

39 225×10

40 $284 \times 1\,000$

41 $68 \times 1\,000$

42 540×10

43 $502 \times 1\,000$

44 437×100

YOU DO

On your own: **Easy** Set B. Evaluate using a strategy.

6 QUESTIONS

1 34×5

4 16×5

2 25×8

5 45×4

3 33×9

6 21×11



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate using the strategy given.

a 34×5 (use $\times 10 \div 2$)

Your teacher works this out. Copy every step.

b 33×9 (use $\times 10 - 1$)

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each product on paper.

6 QUESTIONS

REMEMBER $\times 5$ is the same as $\times 10 \div 2$.

1 126×7

3 375×6

5 1234×3

2 208×4

4 93×8

6 459×9

SKILL DRILL 2 Medium basics: keep practising

32 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 32 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate mentally. Split the number: $34 \times 6 = 30 \times 6 + 4 \times 6$.

- | | | | |
|---------------------------|---------------------------|---------------------------|---------------------------|
| 1 21×9
_____ | 2 45×9
_____ | 3 55×5
_____ | 4 21×6
_____ |
| 5 75×6
_____ | 6 53×4
_____ | 7 16×6
_____ | 8 95×5
_____ |
| 9 61×5
_____ | 10 25×5
_____ | 11 87×4
_____ | 12 54×9
_____ |
| 13 62×5
_____ | 14 52×7
_____ | 15 71×7
_____ | 16 53×5
_____ |
| 17 57×9
_____ | 18 56×7
_____ | 19 68×4
_____ | 20 26×6
_____ |

ROUND 2 Questions 21–32: evaluate on paper.

- | | | |
|----------------------------------|----------------------------------|----------------------------------|
| 21 865×6
<div></div> | 22 941×8
<div></div> | 23 689×9
<div></div> |
| 24 560×4
<div></div> | 25 699×7
<div></div> | 26 870×9
<div></div> |
| 27 353×4
<div></div> | 28 472×4
<div></div> | 29 222×7
<div></div> |
| 30 842×5
<div></div> | 31 538×4
<div></div> | 32 767×8
<div></div> |

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate on paper.

a 126×7

Your teacher works this out. Copy every step.

b 565×18

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate each product on paper.

4 QUESTIONS

REMEMBER $\times 9$ is the same as $\times 10$ – the number.

1 565×18

3 312×45

2 47×23

4 A bus holds 57 people. How many people can 24 buses hold?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate 14×5 .

Level 2

Evaluate 243×6 .

Level 3

Evaluate 128×27 .

Extension · Stretch yourself**Worked example**Find the missing digits: $4\square \times 6 = 2\square 8$

- Units: $\square \times 6$ ends in 8, so $\square = 3$ or 8
- $43 \times 6 = 258$ ✓; $48 \times 6 = 288$ ✓

Answer: $43 \times 6 = 258$ or $48 \times 6 = 288$ **YOU DO**

On your own: read the worked example, then try these.

- E1**
- Find the missing digit:
- $\square 7 \times 5 = 185$
- .

- E2**
- Find two 2-digit numbers whose product is 1000.

Lesson summary

Missed the lesson? Start here.

Remember

- Double and halve: $25 \times 8 = 50 \times 4$.
 3×9 is the same as $\times 10$ – the number.

- $\times 5$ is the same as $\times 10 \div 2$.
- On paper: multiply each digit and carry.

Level 1

12×4

- $10 \times 4 + 2 \times 4$

Answer: 48**Level 2**

34×5

- $340 \div 2$

Answer: 170**Level 3**

565×18

- $565 \times 8 = 4520$
- $565 \times 10 = 5650$
- Add

Answer: 10 170**Extension**

$4\square \times 6 = 2\square 8$

- Try 3 and 8

Answer: 43 or 48

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.03

Division

Date _____

Learning intentions

We are learning to:

- divide whole numbers mentally and using short division

Success criteria

I can:

- ☐ divide by 10, 100 and 1000 mentally
- ☐ use short division with a one-digit divisor
- ☐ write a remainder when the division is not exact

Key terms Write the definition as your teacher explains it.

Quotient

Remainder

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate mentally.

a $2400 \div 10$

b $144 \div 9$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each quotient mentally.

14 QUESTIONS

REMEMBER $\div 10, 100, 1000$: move digits right.

1 $2400 \div 10$

2 $48 \div 6$

3 $72 \div 8$

4 $3500 \div 100$

5 $144 \div 12$

6 $63 \div 9$

Questions 7–14: evaluate mentally.

7 $56 \div 7$

8 $81 \div 9$

9 $120 \div 10$

10 $4500 \div 100$

11 $96 \div 8$

12 $132 \div 11$

13 $64 \div 4$

14 $250 \div 5$

SKILL DRILL 1 Easy basics: practise until it is automatic

44 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 44

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–28: know your division facts.

1 $50 \div 10$

2 $77 \div 11$

3 $84 \div 7$

4 $10 \div 2$

5 $88 \div 11$

6 $42 \div 6$

7 $81 \div 9$

8 $96 \div 12$

9 $54 \div 9$

10 $20 \div 2$

11 $10 \div 5$

12 $14 \div 7$

13 $110 \div 11$

14 $100 \div 10$

15 $30 \div 5$

16 $18 \div 6$

17 $12 \div 2$

18 $45 \div 9$

19 $9 \div 3$

20 $18 \div 3$

21 $24 \div 12$

22 $110 \div 10$

23 $44 \div 11$

24 $32 \div 8$

25 $18 \div 2$

26 $108 \div 12$

27 $36 \div 4$

28 $40 \div 8$

ROUND 2 Questions 29–44: divide by 10, 100 or 1000.

29 $310 \div 10$

30 $12\,000 \div 1\,000$

31 $430 \div 10$

32 $500 \div 10$

33 $9\,400 \div 100$

34 $3\,100 \div 100$

35 $38\,000 \div 1\,000$

36 $90\,000 \div 1\,000$

37 $920 \div 10$

38 $110 \div 10$

39 $64\,000 \div 1\,000$

40 $5\,000 \div 100$

41 $340 \div 10$

42 $290 \div 10$

43 $520 \div 10$

44 $740 \div 10$

YOU DO

On your own: **Easy** Set B. Evaluate each quotient mentally.

6 QUESTIONS

1 $144 \div 9$

4 $96 \div 3$

2 $90 \div 5$

5 $81\,000 \div 1000$

3 $120 \div 4$

6 $84 \div 7$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use short division.

a $620 \div 5$

Your teacher works this out. Copy every step.

b $528 \div 4$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Use short division.

6 QUESTIONS

REMEMBER Short division: divide each digit, carry the remainder.

1 $620 \div 5$

3 $1296 \div 8$

5 $2115 \div 9$

2 $528 \div 4$

4 $847 \div 7$

6 $936 \div 6$

SKILL DRILL 2 Medium basics: keep practising

32 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 32

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate mentally.

1 $210 \div 6$

2 $440 \div 8$

3 $54 \div 2$

4 $282 \div 6$

5 $98 \div 2$

6 $120 \div 8$

7 $123 \div 3$

8 $448 \div 8$

9 $168 \div 6$

10 $392 \div 7$

11 $288 \div 9$

12 $136 \div 4$

13 $232 \div 8$

14 $196 \div 7$

15 $235 \div 5$

16 $228 \div 6$

17 $273 \div 7$

18 $141 \div 3$

19 $336 \div 6$

20 $186 \div 6$

ROUND 2 Questions 21–32: use short division. Write any remainder.

21 $388 \div 7$

22 $955 \div 8$

23 $852 \div 9$

24 $569 \div 3$

25 $114 \div 6$

26 $294 \div 3$

27 $566 \div 9$

28 $183 \div 5$

29 $824 \div 5$

30 $647 \div 4$

31 $938 \div 8$

32 $739 \div 5$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use short division.

a $1296 \div 8$

Your teacher works this out. Copy every step.

b $847 \div 7$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Divide. Write any remainder.

4 QUESTIONS

REMEMBER A remainder is what is left over.

1 $475 \div 6$

3 $5000 \div 7$

2 $1003 \div 4$

4 250 students go on buses that hold 45. How many buses are needed?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate $360 \div 10$.

Level 2

Evaluate $738 \div 6$.

Level 3

Evaluate $1250 \div 8$, with remainder.

 Extension · Stretch yourself Worked example

I am a number between 50 and 70. Divided by 4 I leave remainder 1; divided by 5 I leave remainder 3. What am I?

1. Remainder 1 $\div$ 4: 53, 57, 61, 65, 69
2. Remainder 3 $\div$ 5: 53, 58, 63, 68

Answer: 53

YOU DO

On your own: read the worked example, then try these.

- E1** Find a number between 20 and 40 that leaves remainder 2 when divided by 3, 4 and 6.

- E2** $1000 \div 7$: what is the remainder?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. $\div$ 10, 100, 1000: move digits right.
 3. A remainder is what is left over.

2. Short division: divide each digit, carry the remainder.
4. In a problem, decide whether to round up.

 Level 1

$$2400 \div 10$$

1. Move digits 1 place right

Answer: 240

 Level 2

$$528 \div 4$$

1. $5 \div 4 = 1 \text{ r}1$
2. $12 \div 4 = 3$
3. $8 \div 4 = 2$

Answer: 132

 Level 3

$$475 \div 6$$

1. $47 \div 6 = 7 \text{ r}5$
2. $55 \div 6 = 9 \text{ r}1$

Answer: 79 r 1

 Extension

$$\div 4 \text{ r}1 \text{ and } \div 5 \text{ r}3$$

1. List and compare

Answer: 53

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LESSON
3.04

Divisibility tests

Date _____

Learning intentions

We are learning to:

- use divisibility tests for 2, 3, 4, 5, 6, 8, 9 and 10

Success criteria

I can:

- ☐ test for 2, 5 and 10 using the last digit
- ☐ test for 3 and 9 using the digit sum
- ☐ test for 4, 6 and 8

Key terms Write the definition as your teacher explains it.

Divisible

Digit sum

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the tests for 2, 5 and 10, then for 3 and 9.

a Is 1208 divisible by 2? By 5? By 10?

b Is 1208 divisible by 3? By 9?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write yes or no.

14 QUESTIONS

REMEMBER 2: even. 5: ends in 0 or 5. 10: ends in 0.

1 Is 346 divisible by 2? **2** Is 346 divisible by 5? **3** Is 780 divisible by 10? **4** Is 915 divisible by 5?

5 Is 4071 divisible by 2? **6** Is 5500 divisible by 10?

Questions 7–14: find the digit sum. Is the number divisible by 3?

7 471 **8** 1000 **9** 2226 **10** 5832

11 1208 **12** 999 **13** 4815 **14** 7102

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: is the number divisible by the number in brackets? Write yes or no.

1 8 633 (10)

2 6 686 (2)

3 5 597 (2)

4 2 270 (5)

5 743 (10)

6 2 069 (2)

7 9 467 (2)

8 2 242 (5)

9 2 204 (5)

10 5 018 (5)

11 8 338 (2)

12 9 111 (10)

13 9 893 (5)

14 845 (10)

15 7 667 (2)

16 9 746 (10)

17 1 461 (10)

18 9 818 (10)

19 6 940 (5)

20 4 487 (10)

21 7 491 (10)

22 8 463 (5)

23 3 434 (2)

24 9 304 (5)

ROUND 2 Questions 25–40: find the digit sum. Is the number divisible by 3?

25 2 137

26 3 040

27 9 940

28 7 688

29 2 377

30 571

31 4 853

32 3 220

33 5 427

34 5 847

35 1 804

36 5 496

37 1 144

38 3 307

39 5 911

40 9 577

YOU DO

On your own: **Easy** Set B. Find the digit sum, then answer yes or no.

6 QUESTIONS

1 Is 471 divisible by 3?

4 Is 1000 divisible by 3?

2 Is 471 divisible by 9?

5 Is 2226 divisible by 3?

3 Is 5832 divisible by 9?

6 Is 8109 divisible by 9?



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use the tests for 4, 8 and 6.

a Is 1208 divisible by 4? By 8?

Your teacher works this out. Copy every step.

b Is 1208 divisible by 6?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Test each number for 4, 6 and 8. Show your test.

6 QUESTIONS

REMEMBER 3: digit sum divisible by 3. 9: digit sum by 9.

1 324

3 516

5 2346

2 1208

4 7040

6 9112

SKILL DRILL 2 Medium basics: keep practising

32 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 32 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: divisible by 4? Check the last two digits. Write yes or no.

- | | | | |
|-------------------|-------------------|-------------------|-------------------|
| 1 472
_____ | 2 1 380
_____ | 3 404
_____ | 4 7 896
_____ |
| 5 2 882
_____ | 6 3 392
_____ | 7 3 705
_____ | 8 3 806
_____ |
| 9 6 630
_____ | 10 8 591
_____ | 11 2 393
_____ | 12 7 807
_____ |
| 13 7 400
_____ | 14 289
_____ | 15 3 485
_____ | 16 6 695
_____ |

ROUND 2 Questions 17–32: is the number divisible by 6 (by 2 and 3) or 9 (digit sum)? Write yes or no.

- | | | | |
|-----------------------|-----------------------|-----------------------|-----------------------|
| 17 2 133 (9)
_____ | 18 5 794 (6)
_____ | 19 7 671 (9)
_____ | 20 564 (6)
_____ |
| 21 6 790 (9)
_____ | 22 9 774 (9)
_____ | 23 1 368 (6)
_____ | 24 9 315 (9)
_____ |
| 25 414 (9)
_____ | 26 9 096 (6)
_____ | 27 5 652 (9)
_____ | 28 6 282 (6)
_____ |
| 29 8 226 (9)
_____ | 30 7 777 (6)
_____ | 31 7 736 (9)
_____ | 32 8 100 (9)
_____ |

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use divisibility tests to answer.

- a** Which number is divisible by both 3 and 8: 30, 24, 90 or 94?

Your teacher works this out. Copy every step.

- b** Find the smallest digit $\square$ so that $52\square$ is divisible by 6.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use divisibility tests.

4 QUESTIONS

REMEMBER 4: last 2 digits by 4. 8: last 3 digits by 8.

- 1** Test 2520 for 2, 3, 4, 5, 6, 8, 9 and 10.

- 3** Find all digits $\square$ that make $7\square4$ divisible by 3.

- 2** Which numbers are divisible by both 3 and 8: 30, 24, 90, 96, 120?

- 4** Can a number be divisible by 9 but not by 3? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is 735 divisible by 5?

Level 2

Is 4815 divisible by 9?

Level 3

Is 1716 divisible by 6? Explain.

Extension · Stretch yourself

Worked example

Is 3 162 816 divisible by 12?

- $12 = 3 \times 4$, so test 3 and 4
- Digit sum $3 + 1 + 6 + 2 + 8 + 1 + 6 = 27$, divisible by 3
- Last two digits 16, divisible by 4

Answer: Yes

YOU DO

On your own: read the worked example, then try these.

E1 Is 45 720 divisible by 15? Explain using tests.

E2 Find the smallest 4-digit number divisible by 2, 3, 4, 5 and 6.

Lesson summary

Missed the lesson? Start here.

Divisibility tests

- 2: even. 5: ends in 0 or 5. 10: ends in 0.
- 3: digit sum divisible by 3. 9: digit sum by 9.
- 4: last 2 digits by 4. 8: last 3 digits by 8.
- 6: divisible by 2 and 3.

Level 1

$1208 \div 2?$

- Ends in 8, even

Answer: yes

Level 2

$1208 \div 3?$

- $1 + 2 + 0 + 8 = 11$

Answer: no

Level 3

$1208 \div 8?$

- $208 \div 8 = 26$

Answer: yes

Extension

$3\ 162\ 816 \div 12?$

- Test 3 and 4

Answer: yes

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.05

Long division

Date _____

Learning intentions

We are learning to:

- divide by a two-digit number using long division

Success criteria

I can:

- ☐ set out long division
- ☐ use the steps divide, multiply, subtract, bring down
- ☐ write the answer with a remainder

Key terms Write the definition as your teacher explains it.

Divisor

Dividend

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use long division.

a $650 \div 25$

b $2976 \div 24$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Warm up: know your multiples.

14 QUESTIONS

REMEMBER Divide: how many times does the divisor go in?

1 12×7

2 25×6

3 24×4

4 15×8

5 32×3

6 16×9

Questions 7–14: know your multiples.

7 13×4

8 14×6

9 23×3

10 21×5

11 18×4

12 17×3

13 22×4

14 19×5

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: know your multiples.

1 21×7

2 21×3

3 23×9

4 16×2

5 25×2

6 14×5

7 13×2

8 15×7

9 20×6

10 14×3

11 23×5

12 24×5

13 19×7

14 13×5

15 15×8

16 17×4

17 13×3

18 21×2

19 18×8

20 16×9

21 18×6

22 17×8

23 17×5

24 13×6

ROUND 2 Questions 25–40: how many times does the first number go into the second? Write any remainder.

25 16 into 48

26 14 into 28

27 13 into 66

28 14 into 61

29 11 into 26

30 19 into 186

31 11 into 92

32 24 into 179

33 25 into 116

34 17 into 89

35 17 into 79

36 18 into 105

37 11 into 107

38 19 into 95

39 18 into 137

40 20 into 59

YOU DO

On your own: **Easy** Set B. Evaluate.

6 QUESTIONS

1 $96 \div 12$

4 $240 \div 15$

2 $175 \div 25$

5 $128 \div 16$

3 $144 \div 24$

6 $99 \div 11$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use long division. Write any remainder.

a $3079 \div 12$

Your teacher works this out. Copy every step.

b $864 \div 16$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Use long division.

6 QUESTIONS

REMEMBER Multiply and write it under.

1 $650 \div 25$

3 $864 \div 16$

5 $3825 \div 15$

2 $2976 \div 24$

4 $1344 \div 12$

6 $2574 \div 11$

SKILL DRILL 2 Medium basics: keep practising

15 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 15

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–9: use long division.

1 $1\,218 \div 14$

2 $1\,207 \div 17$

3 $2\,883 \div 31$

4 $1\,940 \div 20$

5 $1\,360 \div 16$

6 $528 \div 12$

7 $1\,590 \div 30$

8 $704 \div 32$

9 $2\,059 \div 29$

ROUND 2 Questions 10–15: use long division. Write the remainder.

10 $533 \div 23$

11 $2\,124 \div 34$

12 $1\,587 \div 27$

13 $444 \div 21$

14 $2\,600 \div 27$

15 $645 \div 14$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use long division, then answer the question.

a $5000 \div 32$

Your teacher works this out. Copy every step.

b How many 45-seat buses are needed for 1000 people?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use long division. Write any remainder.

4 QUESTIONS

REMEMBER Subtract, then bring down the next digit.

1 $3079 \div 12$

3 $9876 \div 21$

2 $5000 \div 32$

4 1000 people need 45-seat buses. How many buses?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate $84 \div 14$.

Level 2

Evaluate $1728 \div 16$.

Level 3

Evaluate $4000 \div 27$ with remainder.

 Extension · Stretch yourself Worked example

A number divided by 23 gives 41 remainder 17.
Find the number.

1. Number = divisor $\times$ quotient + remainder
2. $23 \times 41 + 17 = 943 + 17$

Answer: 960

YOU DO

On your own: read the worked example,
then try these.

- E1** A number divided by 18 gives 55 remainder 4.
Find the number.

- E2** Find the largest 3-digit number that leaves
remainder 5 when divided by 13.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Divide: how many times does the divisor go in?
 2. Multiply and write it under.
 3. Subtract, then bring down the next digit.
 4. Repeat. What is left is the remainder.

 Level 1

$$96 \div 12$$

1. $12 \times 8 = 96$

Answer: 8

 Level 2

$$650 \div 25$$

1. $65 \div 25 = 2 \text{ r}15$
2. $150 \div 25 = 6$

Answer: 26

 Level 3

$$3079 \div 12$$

1. $30 \div 12 = 2 \text{ r}6$
2. $67 \div 12 = 5 \text{ r}7$
3. $79 \div 12 = 6 \text{ r}7$

Answer: 256 r 7

 Extension

$$23 \times 41 + 17$$

1. $943 + 17$

Answer: 960

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.06

Roman numerals

Date _____

Learning intentions

We are learning to:

- read and write Roman numerals

Success criteria

I can:

- ☐ know the values of I, V, X, L, C, D and M
- ☐ add values when a symbol is followed by a smaller or equal one
- ☐ subtract when a smaller symbol comes before a larger one

Key terms Write the definition as your teacher explains it.

Numeral

Symbol

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write each Roman numeral as a number.

a What is XXXVII?

b What is CCXL?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write each Roman numeral as a number.

14 QUESTIONS

REMEMBER I 1, V 5, X 10, L 50, C 100, D 500, M 1000.

1 VIII

2 XII

3 XXXVII

4 LX

5 CCX

6 XV

Questions 7–14: write each number in Roman numerals.

7 3

8 9

9 14

10 19

11 40

12 52

13 99

14 150

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: write each Roman numeral as a number.

1 XCVIII

2 LXI

3 XXXIX

4 XVIII

5 VIII

6 LXXIX

7 LIX

8 XIX

9 LXXXVI

10 XXIII

11 LXVII

12 XLII

13 LXXXIII

14 XCVII

15 LXXXV

16 LXXXII

17 XIII

18 XC

19 XVII

20 LXXXI

21 LXXVI

22 XCII

23 LXXV

24 XXVIII

ROUND 2 Questions 25–40: write each number in Roman numerals.

25 98

26 41

27 85

28 77

29 97

30 75

31 17

32 50

33 26

34 78

35 18

36 82

37 58

38 39

39 43

40 99

YOU DO

On your own: **Easy** Set B. Write each Roman numeral as a number.

6 QUESTIONS

① IV

4 XL

2 IX

5 CCXL

3 XXIV

6 XC

 Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Write as a number, then write in Roman numerals.

a What is XXIV?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

b Write 28 in Roman numerals.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: Medium Set C. Write each number in Roman numerals.

6 QUESTIONS

REMEMBER Larger then smaller: add ($VI = 6$).

① 6

[illegible]

4 74

2 13

[illegible]

5 125

[illegible]

3 28

[illegible]

6 49

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: write each Roman numeral as a number.

1 DCCLXXXVII

2 DCXIX

3 MDCCCXXXIX

4 DCCCII

5 MCMLXXX

6 MCCLXV

7 MCLXXXVI

8 MCCXV

9 DXVIII

10 MCMLXXIV

11 DCXXII

12 MCCCII

13 MCDXLVIII

14 MDCCLXI

15 MXXXVIII

16 CCXLII

ROUND 2 Questions 17–31: write each year in Roman numerals.

17 2003

18 1935

19 1932

20 2013

21 1945

22 1959

23 1993

24 1911

25 2000

26 1995

27 1901

28 1977

29 1966

30 1922

31 1914

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write each number in Roman numerals.

a Write 74 in Roman numerals.

Your teacher works this out. Copy every step.

b Write 125 in Roman numerals.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder numerals.

4 QUESTIONS

REMEMBER Smaller before larger: subtract (IV = 4).

1 Write 1999 in Roman numerals.

3 Write this year and your birth year in Roman numerals.

2 Write MMXXVI as a number.

4 Why is IC not used for 99?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write XVI as a number.

Level 2

Write 39 in Roman numerals.

Level 3

Write 2024 in Roman numerals.

 Extension · Stretch yourself Worked example

Evaluate XLIV + LVI and write the answer in Roman numerals.

1. XLIV = 44, LVI = 56
2. $44 + 56 = 100$

Answer: C

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate CCC – XLV and write the answer in Roman numerals.

- E2** Evaluate XII × IX in Roman numerals.

Lesson summary

Missed the lesson? Start here.

- Roman numerals**
1. I 1, V 5, X 10, L 50, C 100, D 500, M 1000.
 2. Larger then smaller: add (VI = 6).
 3. Smaller before larger: subtract (IV = 4).
 4. Never more than 3 of the same symbol in a row.

 Level 1

XXXVII

1. $10 + 10 + 10 + 5 + 1 + 1$

Answer: 37

 Level 2

CCXL

1. $200 + 40$

Answer: 240

 Level 3

125

1. $100 + 20 + 5$

Answer: CXXV

 Extension

XLIV + LVI

1. $44 + 56 = 100$

Answer: C

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.07

Index notation and powers

Date _____

Learning intentions

We are learning to:

- write repeated multiplication using index notation
- evaluate powers

Success criteria

I can:

- ☐ identify the **base** and **index**
- ☐ write a product like $3 \times 3 \times 3 \times 3$ as 3^4
- ☐ evaluate a power such as 5^3

Key terms Write the definition as your teacher explains it.

Base

Index

Power

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write in index notation.

a Write $3 \times 3 \times 3 \times 3$ in index notation.

Your teacher works this out. Copy every step.

b Write $7 \times 7 \times 7 \times 7 \times 7 \times 7 \times 7$ in index notation.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write in index notation.

14 QUESTIONS

REMEMBER In 5^3 , 5 is the **BASE** and 3 the **INDEX**.

1 $4 \times 4 \times 4$

2 $6 \times 6 \times 6 \times 6 \times 6$

3 9×9

4 $2 \times 2 \times 2 \times 2 \times 2 \times 2$

5 $10 \times 10 \times 10 \times 10$

6 $5 \times 5 \times 5 \times 3 \times 3$

Questions 7–14: evaluate each power.

7 2^3

8 3^2

9 5^2

10 10^3

11 4^2

12 2^5

13 1^8

14 3^3

SKILL DRILL 1 Easy basics: practise until it is automatic

31 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–15: write in index notation.

1 6×6

2 $9 \times 9 \times 9 \times 9 \times 9$

3 $5 \times 5 \times 5 \times 5 \times 5 \times 5$

4 2×2

5 4×4

6 $2 \times 2 \times 2 \times 2 \times 2$

7 $7 \times 7 \times 7$

8 $8 \times 8 \times 8 \times 8$

9 $9 \times 9 \times 9 \times 9 \times 9 \times 9$

10 $7 \times 7 \times 7 \times 7 \times 7 \times 7$

11 $8 \times 8 \times 8$

12 $2 \times 2 \times 2$

13 $4 \times 4 \times 4 \times 4 \times 4 \times 4$

14 $3 \times 3 \times 3$

15 $6 \times 6 \times 6 \times 6$

ROUND 2 Questions 16–31: evaluate each power.

16 6^2

17 7^2

18 3^3

19 10^2

20 9^2

21 2^3

22 8^2

23 5^2

24 4^3

25 3^5

26 3^4

27 2^4

28 5^3

29 3^2

30 4^2

31 2^5

YOU DO

On your own: **Easy** Set B. Write as a product (expanded form).

6 QUESTIONS

1 8^3

4 2^4

2 3^5

5 7^1

3 11^2

6 $2^3 \times 5^2$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate each power.

a Evaluate 6^2 .

Your teacher works this out. Copy every step.

b Evaluate 5^3 .

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate each power.

6 QUESTIONS

REMEMBER $5^3 = 5 \times 5 \times 5$, not 5×3 .

1 6^2

4 10^4

2 5^3

5 3^4

3 2^5

6 1^9

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: write the base and the index, then evaluate.

1 5^2

2 2^4

3 3^2

4 2^3

5 7^2

6 5^4

7 3^4

8 3^3

9 6^3

10 4^4

11 6^2

12 7^3

13 2^2

14 7^4

15 4^2

16 4^3

ROUND 2 Questions 17–31: evaluate. Powers first.

17 $3^3 - 2^2$

18 $3^2 + 4^2$

19 $5^2 + 3^2$

20 $4^2 + 3^2$

21 $2^2 + 3^2$

22 $4^3 + 4^2$

23 $4^2 \times 3^2$

24 $3^3 + 2^2$

25 $5^3 - 3^2$

26 $5^3 \times 2^2$

27 $3^2 + 2^2$

28 $2^3 + 3^2$

29 $3^3 \times 4^2$

30 $5^3 + 4^2$

31 $5^2 - 3^2$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate each power.

a Evaluate 10^5 .

Your teacher works this out. Copy every step.

b Evaluate 2^8 .

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Evaluate.

4 QUESTIONS

REMEMBER Count the equal factors to find the index.

1 2^8

3 $2^3 \times 3^2$

2 $4^3 - 3^3$

4 Which is larger: 2^5 or 5^2 ? Explain.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write $7 \times 7 \times 7$ in index notation.

Level 2

Evaluate 4^3 .

Level 3

Evaluate $3^2 \times 2^4$.

 Extension · Stretch yourself Worked example

Write 64 as a power of 2, 4 and 8.

1. $64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2$
2. $64 = 4 \times 4 \times 4$
3. $64 = 8 \times 8$

Answer: $2^6 = 4^3 = 8^2$

YOU DO

On your own: read the worked example, then try these.

- E1** Write 81 as a power of 3 and of 9.

- E2** Find n if $2^n = 1024$.

Lesson summary

Missed the lesson? Start here.

Remember 1. In 5^3 , 5 is the **base** and 3 the **index**.
3. Count the equal factors to find the index.

2. $5^3 = 5 \times 5 \times 5$, not 5×3 .
4. Squared means index 2; cubed means index 3.

 Level 1

$3 \times 3 \times 3 \times 3$

1. Four 3s

Answer: 3^4

 Level 2

5^3

1. $5 \times 5 \times 5$

Answer: 125

 Level 3

2^8

1. 2, 4, 8, 16, 32, 64, 128, 256

Answer: 256

 Extension

64 as powers

1. 2^6 , 4^3 , 8^2

Answer: 2^6 , 4^3 , 8^2

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.08

Square roots and cube roots

Date _____

Learning intentions

We are learning to:

- find square roots and cube roots

Success criteria

I can:

- ☐ know the square numbers to 15^2 and the cube numbers to 10^3
- ☐ find $\sqrt{\quad}$ as the number that squares to it
- ☐ find $\sqrt[3]{\quad}$ as the number that cubes to it

Key terms Write the definition as your teacher explains it.

Square root

Cube root

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate each square root.

a $\sqrt{49}$

b $\sqrt{144}$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate each square root.

14 QUESTIONS

REMEMBER $\sqrt{49} = 7$ because $7^2 = 49$.

1 $\sqrt{16}$

2 $\sqrt{49}$

3 $\sqrt{81}$

4 $\sqrt{100}$

5 $\sqrt{144}$

6 $\sqrt{1}$

Questions 7–14: evaluate.

7 $\sqrt{25}$

8 $\sqrt{64}$

9 $\sqrt{121}$

10 $\sqrt{400}$

11 $\sqrt[3]{27}$

12 $\sqrt[3]{64}$

13 $\sqrt[3]{125}$

14 $\sqrt[3]{1000}$

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate each square root.

1 $\sqrt{4}$

2 $\sqrt{36}$

3 $\sqrt{484}$

4 $\sqrt{196}$

5 $\sqrt{400}$

6 $\sqrt{289}$

7 $\sqrt{441}$

8 $\sqrt{225}$

9 $\sqrt{144}$

10 $\sqrt{576}$

11 $\sqrt{625}$

12 $\sqrt{1}$

13 $\sqrt{169}$

14 $\sqrt{9}$

15 $\sqrt{25}$

16 $\sqrt{361}$

17 $\sqrt{81}$

18 $\sqrt{49}$

19 $\sqrt{529}$

20 $\sqrt{256}$

ROUND 2 Questions 21–36: evaluate each square.

21 17^2

22 20^2

23 7^2

24 6^2

25 2^2

26 1^2

27 3^2

28 12^2

29 5^2

30 9^2

31 10^2

32 16^2

33 11^2

34 13^2

35 14^2

36 15^2

YOU DO

On your own: **Easy** Set B. Evaluate each cube root.

6 QUESTIONS

1 $\sqrt[3]{8}$

4 $\sqrt[3]{125}$

2 $\sqrt[3]{27}$

5 $\sqrt[3]{1000}$

3 $\sqrt[3]{64}$

6 $\sqrt[3]{1}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate each cube root.

a $\sqrt[3]{8000}$

Your teacher works this out. Copy every step.

b $\sqrt[3]{729}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate.

6 QUESTIONS

REMEMBER $\sqrt[3]{64} = 4$ because $4^3 = 64$.

1 $\sqrt{400}$

4 $\sqrt{225}$

2 $\sqrt[3]{8000}$

5 $\sqrt{36} + \sqrt{64}$

3 $\sqrt[3]{729}$

6 $\sqrt[3]{216} \times \sqrt{4}$

SKILL DRILL 2 Medium basics: keep practising

35 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 35

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: evaluate each cube root or cube.

1 $\sqrt[3]{8}$

2 9^3

3 6^3

4 5^3

5 4^3

6 $\sqrt[3]{729}$

7 2^3

8 7^3

9 10^3

10 3^3

11 $\sqrt[3]{27}$

12 $\sqrt[3]{125}$

13 $\sqrt[3]{512}$

14 $\sqrt[3]{1000}$

15 $\sqrt[3]{64}$

16 $\sqrt[3]{1}$

17 $\sqrt[3]{343}$

18 8^3

19 $\sqrt[3]{216}$

20 1^3

ROUND 2 Questions 21–35: between which two whole numbers is it?

21 $\sqrt{60}$

22 $\sqrt{52}$

23 $\sqrt{116}$

24 $\sqrt{145}$

25 $\sqrt{17}$

26 $\sqrt{150}$

27 $\sqrt{33}$

28 $\sqrt{24}$

29 $\sqrt{68}$

30 $\sqrt{85}$

31 $\sqrt{128}$

32 $\sqrt{107}$

33 $\sqrt{141}$

34 $\sqrt{57}$

35 $\sqrt{148}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate, then estimate.

a $\sqrt{(9 \times 16)}$

Your teacher works this out. Copy every step.

b Between which two whole numbers is $\sqrt{50}$?

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Learn the squares to 15^2 and cubes to 10^3 .

1 Between which two whole numbers is $\sqrt{90}$?

3 Evaluate $\sqrt{(36 + 64)}$. Is it the same as $\sqrt{36} + \sqrt{64}$?

2 Between which two whole numbers is $\sqrt[3]{100}$?

4 A square has area 196 cm^2 . Find its side length.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate $\sqrt{64}$.

Level 2

Evaluate $\sqrt[3]{125}$.

Level 3

Evaluate $\sqrt{121} - \sqrt[3]{1000}$.

Extension · Stretch yourself

Worked example

Find $\sqrt{1600}$ without a calculator.

1. $1600 = 16 \times 100$
2. $\sqrt{16} = 4$ and $\sqrt{100} = 10$

Answer: 40

YOU DO

On your own: read the worked example, then try these.

- E1 Find $\sqrt{4900}$ and $\sqrt[3]{27\,000}$ without a calculator.

- E2 A cube has volume 343 cm^3 . Find the length of its edge.

Lesson summary

Missed the lesson? Start here.

Remember

1. $\sqrt{49} = 7$ because $7^2 = 49$.
2. $\sqrt[3]{64} = 4$ because $4^3 = 64$.
3. Learn the squares to 15^2 and cubes to 10^3 .
4. Estimate a root using the nearest squares.

Level 1

$\sqrt{144}$

1. $12 \times 12 = 144$

Answer: 12

Level 2

$\sqrt[3]{729}$

1. $9 \times 9 \times 9 = 729$

Answer: 9

Level 3

$\sqrt{50}$

1. $49 < 50 < 64$

Answer: between 7 and 8

Extension

$\sqrt{1600}$

1. 4×10

Answer: 40

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.09

Prime and composite numbers

Date _____

Learning intentions

We are learning to:

- identify prime and composite numbers

Success criteria

I can:

- ☐ know a prime has exactly 2 factors
- ☐ know a composite number has more than 2 factors
- ☐ list the primes and composites in a range

Key terms Write the definition as your teacher explains it.

Factor

Prime

Composite

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 List the factors, then decide.

a List the factors of 12 and of 13.

b Is 1 prime, composite or neither?

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: List all the factors of each number.

14 QUESTIONS

REMEMBER A **PRIME** has exactly 2 factors: 1 and itself.

1 6

2 11

3 15

4 20

5 17

6 36

Questions 7–14: prime (P) or composite (C)?

7 13

8 21

9 29

10 33

11 41

12 49

13 57

14 61

SKILL DRILL 1 Easy basics: practise until it is automatic

40 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 40

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–28: prime (P) or composite (C)?

1 92

2 18

3 86

4 16

5 33

6 78

7 99

8 79

9 75

10 65

11 13

12 52

13 70

14 97

15 21

16 38

17 31

18 68

19 30

20 48

21 67

22 45

23 8

24 22

25 23

26 56

27 54

28 90

ROUND 2 Questions 29–40: list all the factors.

29 24

30 13

31 25

32 31

33 41

34 12

35 35

36 45

37 44

38 46

39 9

40 42

YOU DO

On your own: **Easy** Set B. Prime or composite?

6 QUESTIONS

1 7

4 39

2 21

5 47

3 29

6 91



Level 2 · Building up

Medium · 4 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

List the primes, then decide.

a List the prime numbers less than 30.

Your teacher works this out. Copy every step.

b Is 51 prime? Explain.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. List the numbers.

4 QUESTIONS

REMEMBER A **COMPOSITE** number has more than 2 factors.

1 The prime numbers between 50 and 70.

3 The composite numbers larger than 40 but less than 59.

2 The composite numbers between 20 and 32.

4 The composite numbers between 45 and 70 that are odd.

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: how many factors does the number have?

1 81

2 65

3 29

4 69

5 60

6 73

7 88

8 40

9 79

10 26

11 75

12 62

13 46

14 21

15 49

16 28

ROUND 2 Questions 17–31: prime? Write P. If not, write a factor pair (other than 1 and itself).

17 131

18 92

19 99

20 64

21 149

22 128

23 133

24 147

25 157

26 72

27 121

28 113

29 154

30 120

31 179

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 List the numbers.

a List the prime numbers between 26 and 50.

Your teacher works this out. Copy every step.

b List the composite numbers between 45 and 60.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER 1 is neither prime nor composite.

1 Why is 2 the only even prime number?

3 Twin primes differ by 2, like 11 and 13. Find all twin primes under 50.

2 Find two primes that add to 36 (find 2 ways).

4 Is $2 \times 3 \times 5 + 1$ prime?

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is 23 prime or composite?

Level 2

List the factors of 28.

Level 3

List the primes between 30 and 45.

 Extension · Stretch yourself Worked example

Is 143 prime?

1. Only test primes up to $\sqrt{143} \approx 12$: 2, 3, 5, 7, 11
2. $143 \div 11 = 13$

Answer: No: $143 = 11 \times 13$

YOU DO

On your own: read the worked example, then try these.

E1 Is 197 prime? Show your tests.

E2 Is 221 prime?

Lesson summary

Missed the lesson? Start here.

Remember

1. A **prime** has exactly 2 factors: 1 and itself.
2. A **composite** number has more than 2 factors.
3. 1 is neither prime nor composite.
4. Test for primes by dividing by 2, 3, 5, 7, ...

 Level 1

Factors of 12

1. 1×12 , 2×6 , 3×4

Answer: 1, 2, 3, 4, 6, 12

 Level 2

Is 51 prime?

1. $5 + 1 = 6$ so divisible by 3

Answer: no, $51 = 3 \times 17$

 Level 3

Primes 26 to 50

1. Test each odd number

Answer: 29, 31, 37, 41, 43, 47

 Extension

Is 143 prime?

1. Try 11

Answer: no, 11×13

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.10

Prime factors

Date _____

Learning intentions

We are learning to:

- write a number as a product of prime factors using a factor tree

Success criteria

I can:

- ☐ draw a factor tree until every branch ends in a prime
- ☐ write the answer in index notation
- ☐ use prime factors to find square and cube roots

Key terms Write the definition as your teacher explains it.

Factor tree

Prime factor

Notes

Level 1 · Getting started

Easy · 17 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Draw a factor tree for each number.

a Use a factor tree for 60.

b Use a factor tree for 200.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Draw a factor tree for each number.

6 QUESTIONS

REMEMBER Split the number into two factors.

1 12

3 30

5 45

2 18

4 40

6 50

YOU DO

On your own: **Easy** Set A2. Quick drill: write as a product of two prime numbers.

8 QUESTIONS

1 6

2 15

3 21

4 35

5 22

6 39

7 26

8 77

YOU DO

On your own: **Easy** Set B. Write each number as a product of primes in index notation.

3 QUESTIONS

1 24

2 36

3 72

SKILL DRILL 1 Easy basics: practise until it is automatic

32 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 32 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–20: write as a product of two prime numbers.

- | | | | |
|-----------------|-----------------|----------------|----------------|
| 1 26
_____ | 2 65
_____ | 3 39
_____ | 4 22
_____ |
| 5 49
_____ | 6 121
_____ | 7 9
_____ | 8 91
_____ |
| 9 14
_____ | 10 35
_____ | 11 33
_____ | 12 4
_____ |
| 13 169
_____ | 14 143
_____ | 15 25
_____ | 16 15
_____ |
| 17 55
_____ | 18 10
_____ | 19 21
_____ | 20 6
_____ |

ROUND 2 Questions 21–32: write as a product of prime factors in index notation.

- | | | |
|----------------|----------------|----------------|
| 21 20
_____ | 22 8
_____ | 23 66
_____ |
| 24 46
_____ | 25 28
_____ | 26 45
_____ |
| 27 57
_____ | 28 72
_____ | 29 15
_____ |
| 30 54
_____ | 31 24
_____ | 32 70
_____ |

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Write each number as a product of primes in index notation.

a Use a factor tree for 144.

Your teacher works this out. Copy every step.

b Write 90 as a product of primes in index notation.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use a factor tree. Write in index notation.

6 QUESTIONS

REMEMBER Keep splitting until every branch is prime.

1 60

3 144

5 675

2 200

4 252

6 392

SKILL DRILL 2 Medium basics: keep practising

25 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line (show working in the box).

Start time _____ Finish time _____ Score _____ / 25 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1

Questions 1–9: draw a factor tree, then write the number as a product of primes in index notation.

1 280

2 306

3 339

4 187

5 284

6 391

7 384

8 162

9 361

ROUND 2 Questions 10–25: evaluate.

10 2×5

11 $2 \times 3 \times 5$

12 $2^3 \times 5$

13 $2^2 \times 5$

14 2

15 $2^2 \times 3^2$

16 2×3^2

17 $2^2 \times 3$

18 $2 \times 3^2 \times 5$

19 2^2

20 $2^3 \times 3^2$

21 $2^2 \times 3 \times 5$

22 $2^3 \times 3$

23 $2^3 \times 3 \times 5$

24 2×3

25 $2^3 \times 3^2 \times 5$

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use prime factors to find each root.

a Use a factor tree to find $\sqrt{256}$.

Your teacher works this out. Copy every step.

b Use a factor tree to find $\sqrt[3]{5832}$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Use prime factors to evaluate each root.

4 QUESTIONS

REMEMBER Write the primes in order, using indices.

1 $\sqrt{256}$

3 $\sqrt[3]{5832}$

2 $\sqrt{324}$

4 $\sqrt[3]{2744}$

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

What is the smallest number that 72 must be multiplied by to make a perfect square?

1. $72 = 2^3 \times 3^2$
2. A square needs even indices
3. $\times 2$ gives $2^4 \times 3^2 = 144$

Answer: 2

YOU DO

On your own: read the worked example, then try these.

- E1** Find the smallest number that 180 must be multiplied by to make a perfect square.

- E2** Find the smallest number that 18 must be multiplied by to make a perfect cube.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write 20 as a product of primes.

Level 2

Write 84 as a product of primes in index notation.

Level 3

Use prime factors to find $\sqrt{576}$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Split the number into two factors.
2. Keep splitting until every branch is prime.
3. Write the primes in order, using indices.
4. For $\sqrt{\quad}$, halve the indices; for $\sqrt[3]{\quad}$, divide them by 3.

Level 1

Factor tree for 60

1. $60 = 6 \times 10 = 2 \times 3 \times 2 \times 5$

Answer: $2^2 \times 3 \times 5$

Level 2

200

1. $200 = 2 \times 100 = 2 \times 10 \times 10$

Answer: $2^3 \times 5^2$

Level 3

$\sqrt{256}$

1. $256 = 2^8$
2. Halve the index: 2^4

Answer: 16

Extension

$72 \times ? = \text{square}$

1. $2^3 \times 3^2$

Answer: 2

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.11

Highest common factor

Date _____

Learning intentions

We are learning to:

- find the highest common factor (HCF) of two numbers

Success criteria

I can:

- ☐ list the factors of each number and find the largest common one
- ☐ use prime factors to find the HCF
- ☐ use the HCF to solve problems

Key terms Write the definition as your teacher explains it.

Common factor

HCF

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the HCF by listing factors.

a Find the HCF of 12 and 18.

b Find the HCF of 20 and 48.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Find the HCF by listing factors.

14 QUESTIONS

REMEMBER List the factors of both numbers.

1 6 and 9

2 8 and 12

3 10 and 15

4 14 and 21

5 16 and 24

6 9 and 20

Questions 7–14: find the HCF.

7 4 and 6

8 9 and 12

9 10 and 25

10 8 and 20

11 18 and 24

12 7 and 21

13 15 and 40

14 12 and 30

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO

On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find the highest common factor (HCF).

1 30 and 15
_____2 12 and 28
_____3 20 and 12
_____4 14 and 2
_____5 14 and 35
_____6 15 and 35
_____7 30 and 36
_____8 18 and 45
_____9 35 and 28
_____10 5 and 25
_____11 7 and 35
_____12 30 and 6
_____13 5 and 15
_____14 15 and 21
_____15 24 and 42
_____16 56 and 32
_____17 12 and 6
_____18 12 and 3
_____19 15 and 10
_____20 24 and 4
_____21 42 and 18
_____22 9 and 18
_____23 20 and 16
_____24 16 and 24

ROUND 2 Questions 25–36: list the common factors.

25 8 and 12
_____26 4 and 12
_____27 16 and 8
_____28 15 and 18
_____29 24 and 16
_____30 10 and 20
_____31 10 and 25
_____32 15 and 30
_____33 25 and 30
_____34 20 and 10
_____35 12 and 6
_____36 10 and 12

YOU DO

On your own: **Easy** Set B. Find the HCF.

6 QUESTIONS

1 20 and 48

4 24 and 40

2 36 and 84

5 18 and 54

3 45 and 60

6 28 and 42



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the HCF.

a Find the HCF of 36 and 84.

Your teacher works this out. Copy every step.

b Find the HCF of 17 and 34.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Use factor trees to find the HCF.

6 QUESTIONS

REMEMBER Circle the common factors.

1 243 and 162

3 96 and 144

5 150 and 225

2 120 and 200

4 84 and 126

6 98 and 140

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: find the HCF.

1 36 and 72
_____2 18 and 63
_____3 30 and 42
_____4 40 and 72
_____5 112 and 56
_____6 20 and 30
_____7 144 and 54
_____8 18 and 72
_____9 144 and 80
_____10 44 and 55
_____11 8 and 28
_____12 28 and 63
_____13 63 and 35
_____14 72 and 96
_____15 60 and 120
_____16 96 and 64

ROUND 2 Questions 17–31: find the HCF of the three numbers.

17 24, 40 and 8
_____18 2, 12 and 6
_____19 45, 9 and 27
_____20 20, 60 and 30
_____21 10, 6 and 8
_____22 15, 30 and 10
_____23 24, 36 and 72
_____24 24, 32 and 16
_____25 48, 32 and 24
_____26 50, 40 and 10
_____27 22, 44 and 11
_____28 7, 21 and 42
_____29 12, 16 and 24
_____30 40, 50 and 30
_____31 66, 33 and 11

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use factor trees to find the HCF.

a Use factor trees to find the HCF of 243 and 162.

Your teacher works this out. Copy every step.

b Use factor trees to find the HCF of 180 and 252.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. HCF problems.

4 QUESTIONS

REMEMBER The largest one is the HCF.

1 Simplify $36/84$ using the HCF.

3 Two ribbons are 45 cm and 75 cm long. They are cut into equal pieces as long as possible. How long is each piece?

2 24 boys and 32 girls are put into equal teams with no mixing. What is the largest team size?

4 The HCF of two numbers is 6. Both are between 20 and 40. Find a possible pair.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the HCF of 8 and 20.

Level 2

Find the HCF of 42 and 70.

Level 3

Use prime factors to find the HCF of 168 and 252.

 Extension · Stretch yourself Worked example

Find the HCF of 36, 60 and 84.

- $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$
- Common primes: $2^2 \times 3$

Answer: 12

YOU DO

On your own: read the worked example, then try these.

- E1** Find the HCF of 48, 72 and 120.

- E2** Find the HCF of 105, 165 and 195.

Lesson summary

Missed the lesson? Start here.

Remember 1. List the factors of both numbers.
3. The largest one is the **HCF**.

2. Circle the common factors.
4. Or: multiply the prime factors they share.

 Level 1

HCF of 12 and 18

1. 12: 1, 2, 3, 4, 6, 12
2. 18: 1, 2, 3, 6, 9, 18

Answer: 6

 Level 2

HCF of 20 and 48

1. Common: 1, 2, 4

Answer: 4

 Level 3

HCF of 243 and 162

1. $243 = 3^5$
2. $162 = 2 \times 3^4$

Answer: $3^4 = 81$

 Extension

HCF of 36, 60, 84

1. $2^2 \times 3$

Answer: 12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
3.12

Lowest common multiple

Date _____

Learning intentions

We are learning to:

- find the lowest common multiple (LCM) of two numbers

Success criteria

I can:

- ☐ list multiples of each number and find the smallest common one
- ☐ use prime factors to find the LCM
- ☐ use the LCM to solve problems

Key terms Write the definition as your teacher explains it.

Multiple

LCM

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find the LCM by listing multiples.

a Find the LCM of 4 and 6.

b Find the LCM of 6 and 8.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: List the first 6 multiples of each number.

14 QUESTIONS

REMEMBER List multiples of each number.

1 3

2 5

3 8

4 12

5 9

6 15

Questions 7–14: find the LCM.

7 2 and 5

8 3 and 4

9 4 and 10

10 6 and 9

11 5 and 8

12 6 and 10

13 8 and 12

14 9 and 12

SKILL DRILL 1 Easy basics: practise until it is automatic

36 questions

YOU DO On your own: **Easy** Skill drill 1. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 36 Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–24: find the lowest common multiple (LCM).

- | | | | |
|-----------------------|----------------------|-----------------------|----------------------|
| 1 8 and 9
_____ | 2 3 and 12
_____ | 3 7 and 8
_____ | 4 11 and 2
_____ |
| 5 9 and 3
_____ | 6 5 and 10
_____ | 7 3 and 6
_____ | 8 5 and 12
_____ |
| 9 10 and 5
_____ | 10 8 and 7
_____ | 11 5 and 3
_____ | 12 4 and 12
_____ |
| 13 10 and 12
_____ | 14 2 and 9
_____ | 15 10 and 6
_____ | 16 6 and 7
_____ |
| 17 10 and 8
_____ | 18 9 and 5
_____ | 19 5 and 7
_____ | 20 12 and 4
_____ |
| 21 5 and 11
_____ | 22 12 and 5
_____ | 23 11 and 12
_____ | 24 12 and 8
_____ |

ROUND 2 Questions 25–36: write the first three common multiples.

- | | | |
|---------------------|---------------------|---------------------|
| 25 4 and 8
_____ | 26 3 and 5
_____ | 27 3 and 6
_____ |
| 28 6 and 7
_____ | 29 7 and 3
_____ | 30 3 and 8
_____ |
| 31 2 and 5
_____ | 32 4 and 2
_____ | 33 8 and 2
_____ |
| 34 5 and 4
_____ | 35 3 and 7
_____ | 36 2 and 6
_____ |

YOU DO

On your own: **Easy** Set B. Find the LCM.

6 QUESTIONS

1 2 and 3

4 3 and 7

2 4 and 10

5 8 and 12

3 6 and 9

6 10 and 15



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Find the LCM.

a Find the LCM of 18 and 10.

Your teacher works this out. Copy every step.

b Find the LCM of 5 and 7.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Find the LCM. Use factor trees for larger numbers.

6 QUESTIONS

REMEMBER The smallest common one is the LCM.

1 18 and 10

3 12 and 90

5 24 and 36

2 25 and 16

4 14 and 21

6 20 and 45

SKILL DRILL 2 Medium basics: keep practising

31 questions

YOU DO

On your own: **Medium** Skill drill 2. Work down the page, one round at a time. Write each answer on its line.

Start time _____ Finish time _____ Score _____ / 31

Mark it with your teacher. Aim to beat your time next lesson.

ROUND 1 Questions 1–16: find the LCM.

1 20 and 10

2 15 and 9

3 22 and 6

4 15 and 17

5 11 and 12

6 7 and 14

7 8 and 10

8 17 and 21

9 21 and 15

10 8 and 7

11 21 and 19

12 12 and 10

13 24 and 13

14 12 and 6

15 17 and 13

16 9 and 10

ROUND 2 Questions 17–31: find the LCM of the three numbers.

17 8, 6 and 2

18 5, 4 and 7

19 5, 9 and 8

20 10, 4 and 3

21 4, 2 and 5

22 10, 2 and 5

23 8, 9 and 2

24 7, 3 and 4

25 2, 7 and 5

26 10, 5 and 6

27 2, 9 and 8

28 8, 2 and 4

29 7, 3 and 5

30 9, 10 and 3

31 9, 10 and 4

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Use factor trees to find the LCM.

a Use factor trees to find the LCM of 25 and 16.

Your teacher works this out. Copy every step.

b Use factor trees to find the LCM of 12 and 90.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. LCM problems.

4 QUESTIONS

REMEMBER Prime factors: take the highest power of each prime.

1 Buses leave at 6:00 am. One route runs every 12 minutes and another every 15 minutes. When do they next leave together?

3 Hot dogs come in packs of 10 and buns in packs of 8. What is the smallest number of each you can buy to have no leftovers?

2 Find the lowest common denominator of $\frac{5}{6}$ and $\frac{3}{8}$.

4 The LCM of two numbers is 24 and one number is 8. Find all possible values of the other number.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the LCM of 3 and 4.

Level 2

Find the LCM of 12 and 16.

Level 3

Use prime factors to find the LCM of 28 and 42.

 Extension · Stretch yourself Worked example

Find the LCM of 4, 6 and 10.

1. $4 = 2^2$, $6 = 2 \times 3$, $10 = 2 \times 5$
2. Highest power of each prime: $2^2 \times 3 \times 5$

Answer: 60

YOU DO

On your own: read the worked example, then try these.

- E1** Find the LCM of 6, 8 and 9.

- E2** Three lights flash every 4, 5 and 6 seconds. They flash together now. When do they next flash together?

Lesson summary

Missed the lesson? Start here.

Remember

1. List multiples of each number.
2. The smallest common one is the **LCM**.
3. Prime factors: take the highest power of each prime.
4. $\text{LCM} \times \text{HCF} = \text{the product of the two numbers.}$

 Level 1

LCM of 4 and 6

1. 4, 8, 12
2. 6, 12

Answer: 12

 Level 2

LCM of 18 and 10

1. 18, 36, 54, 72, 90
2. 10, ..., 90

Answer: 90

 Level 3

LCM of 25 and 16

1. 5^2 and 2^4
2. No common primes: multiply

Answer: 400

 Extension

LCM of 4, 6, 10

1. $2^2 \times 3 \times 5$

Answer: 60

This space is left blank on purpose. It is the back of your exit ticket.